LINEAR SYSTEMS · THE ALGORITHM

Gaussian elimination, stage by stage

Two stages are compulsory and the third is a choice. Forward elimination and classification happen once; after that, back substitution and Gauss–Jordan reach the same solution set by different amounts of work.

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STAGES

COMPULSORY — EVERY SYSTEM, EVERY TIME 2

1 **Forward elimination**

swaps and row additions, left to right

$[A \mid \mathbf{b}] \to$ row echelon form

Sweep column by column, using each pivot to zero everything beneath it. This is the expensive stage at roughly $\frac{2}{3}n^3$ operations, and it is where **partial pivoting** belongs — choosing the largest available pivot rather than the first nonzero one.

2 **Classify the columns**

and check for $[0 \cdots 0 \mid d \neq 0]$

pivot columns vs free columns

Everything diagnostic happens here. The pivot count is the **rank**, the free columns are the parameters, and a contradiction row settles **solvability** before any solution is attempted.

A CHOICE — TWO WAYS TO READ THE ANSWER 2

3 **Back substitution**

stops at REF

solve upward from the last pivot row

Less arithmetic: roughly n^2 operations on top of the elimination, and no further row operations at all. The usual choice by hand and the one every numerical library uses, since the triangular form is already enough to solve from.

4 **Gauss–Jordan**

zero above the pivots, scale pivots to 1

continue to reduced row echelon form

Roughly 50% more arithmetic than back substitution for the same solution, so it is not the method of choice for solving. Its value is elsewhere: the RREF is unique, which makes it the right form for reading a **null space basis** or inverting a matrix.

WHAT COMES OUT 1

5 **The solution set**

decided at the classification stage

a point, or $\mathbf{x}_p + \mathrm{Null}(A)$, or empty

Never anything else. Unique when every column has a pivot, a translated subspace when some do not, empty when a contradiction row appeared — and which of the three it is was known before the solution was computed.

The whole algorithm is three **row operations** applied until the shape is triangular, and every one of them is reversible — which is why the solution set is unchanged at every step. Elimination does not solve the system so much as rewrite it into a form where the solution can be read.